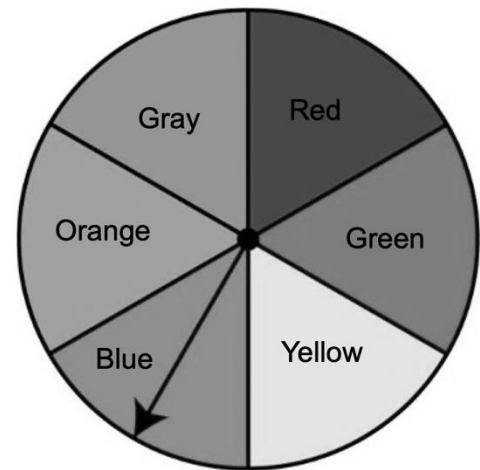


1. Find the theoretical probability of landing on each color as a percentage rounded to the nearest tenth.

Color	Theoretical Probability
Blue	$\frac{1}{6} = 16.7\%$
Yellow	16.7%
Green	16.7%
Red	16.7%
Gray	16.7%
Orange	16.7%



A computer simulator spinner performed more trials. The results as percentages are shown in the table below.

Color	Simulation 1 (10 trials)	Simulation 2 (100 trials)	Simulation 3 (1,000 trials)
Blue	10%	19.1%	16.7%
Yellow	30%	17.3%	16.7%
Green	10%	14.5%	16.7%
Red	20%	12.7%	16.7%
Gray	20%	13.6%	16.7%
Orange	10%	22.7%	16.7%

2. What do you notice in comparing Simulation 1 to Simulation 2?
With more simulations, the probabilities for each color get closer to the theoretical probability.
3. Complete the table by estimating the percentages for Simulation 3.
They should be closer to the theoretical probability the more times the spinner is spun.

Your teacher gives you a standard number cube with numbers 1 – 6 on its faces. You roll it 500 times. Your results are shown in the table below.

Outcome	1	2	3	4	5	6
Frequency	56	92	80	95	67	80

4. What is the experimental probability of rolling an odd number?

$$\frac{56+80+67}{500} = \frac{203}{500} = 0.406 \text{ or } 40.6\%$$

5. How many times would you expect to roll an odd number?

$$\frac{3}{6} = \frac{1}{2} \text{ or } 50\% \text{ of the time.}$$

6. As the number of trials in this simple experiment increases, the
gets closer to the

- experimental probability.
- theoretical probability.

- experimental probability
- theoretical probability

Four situations are described. For each situation, create a possible simulation to fairly represent it.

Answers may vary.

7. Suppose that, on average, a basketball player makes about three out of every four foul shots.
Flipping a coin where landing on heads represents making the foul shot, and tails represent missing it.
8. In three of the past five years, New York has had major snowstorms in March.
Flipping a coin could be used again. Heads means snowstorm in March and tails means no snowstorm. Flip the coin 5 times.
9. Your school is taking 6 buses of students on a field trip. Will you be assigned to the same bus that your science teacher is riding on?
Rolling a fair six-sided die to assign people to 1 of the 6 buses could be used.
10. An experiment will produce one of three different outcomes with equal probability for each. Why would using a coin to simulate the experiment be a bad choice?
Because a coin only has 2 outcomes but the experiment has 3. A fair six-sided die would be better because: rolling a 1 or 2 (first outcome); rolling a 3 or 4 (second outcome); rolling a 5 or 6 (third outcome).